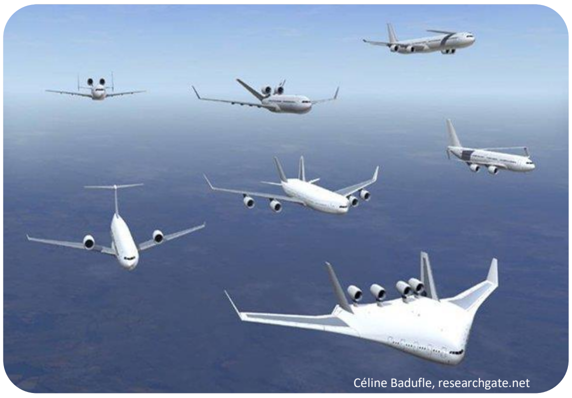
Tutorial VIII – Configurations & Design Process

For aircraft designers it is essential to understand how design choices affect the resulting aircraft properties. With this knowledge the designer is able to design for requirements. This exercise shows how aircraft properties can be traced back to configurative features by taking the example of three civil aircraft.


Céline Badufle, researchgate.net

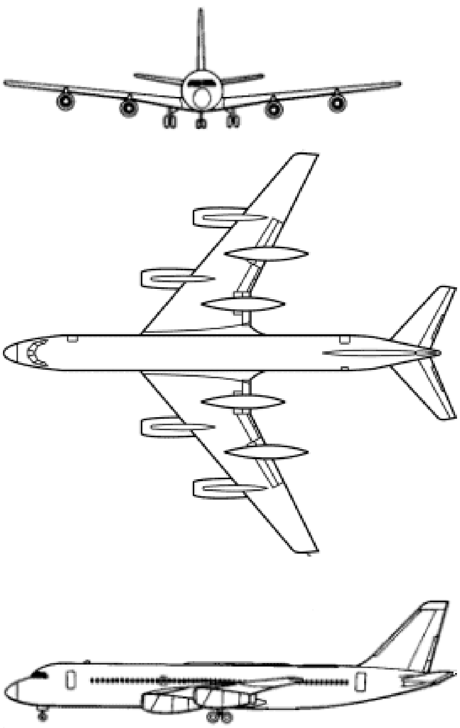
Basics

- 🕮 Explain the aircraft design process. What disciplines take part in the process?
- 🕮 How would you proceed when deciding on the configuration of a new aircraft?
- 🕮 List some customer requirements/ design drivers for new aircraft.
- 🕮 What are advantages and disadvantages of a V-tail and a cruciform tail?

Tasks

Which **configurative features** are visible on the three-side view and the illustration of each aircraft? What are the corresponding **derived properties** for the aircraft? List 10 for each aircraft.

1) Convair 990: Mach 0.92 long-haul passenger jet of the 1960s

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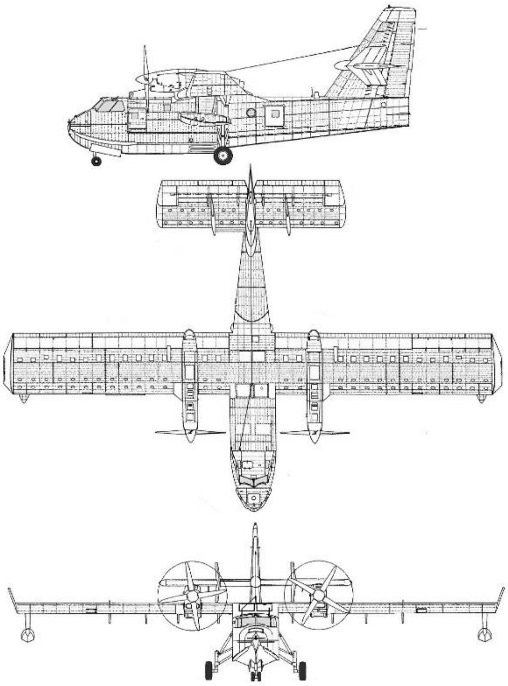
2) Canadair CL-415: Low speed, amphibious firefighting aircraft



3) Learjet 60: Mach 0.74 medium-haul business jet